

Education

Carnegie Mellon University , Pittsburgh, PA	2024
PhD candidate in Computer Science. Advised by Professor Deva Ramanan.	(expected)
Stanford University , Stanford, CA	06/18
MS in Computer Science.	
Stanford University , Stanford, CA	06/12
BS in Computer Science , minor in Management Science and Engineering.	

Professional Experience

Research Scientist Intern , Meta, Pittsburgh, PA	05/23-Present
- Exploring scalable 3D neural rendering topics with Drs. Christian Richardt and Michael Zollhoefer	
Research Intern , Argo AI, Pittsburgh, PA	05/22-08/22
- Explored city-scale 3D reconstruction methods with Professor Deva Ramanan and Dr. Francesco Ferroni	
Engineering Manager and Technical Lead , Palantir Technologies, New York, NY	08/12-08/19
- Led R&D efforts to improve Palantir's Foundry data platform: - o Created interactive search and analysis infrastructure that evolved into one of the primary user interfaces of Palantir's Foundry data lake platform deployed across the company's customer fleet - o Invented full-text search system that powers Palantir's internal log analysis infrastructure and a significant part of the Foundry data lake platform - Led inception and rollout of one of Palantir's flagship data analytics products - Managed the technical implementation of a variety of high-profile customer engagements in the commercial sector (leading teams of 10-100+ individuals) Supervised and mentored a team of 7+ direct reports	
Cofounder , TradeSensei, Palo Alto, CA	01/12-08/12
- Founded a trading startup and wrote the backend infrastructure enabling beta users to participate in real-time trading competitions - Participated in Stanford StartX accelerator program	
Software Engineering Intern , Addepar, Mountain View, CA	04/11-09/11
- Worked on developing a software platform to help private financial advisors and family offices perform quantitative analysis and interface with their clients	
Research Assistant , Stanford University, Stanford, CA	06/10-12/10
- Conducted computer vision research under Professor Daphne Koller - Co-authored two accepted conference papers	

Teaching Experience

Graduate Student Instructor, Dept. of Computer Science , Carnegie Mellon University	09/20-Present
- Teaching assistant for 15-640 (Distributed Systems) and inaugural 17-700 class (Data Science and Machine Learning at Scale) - Supported and mentored team of undergraduate teaching assistants	

- Created and graded assignments, exams, and class projects
- Designed and presented lecture materials
- Supported students during office hours and class forums

Project Mentor, Dept. of Computer Science, Carnegie Mellon University

09/20-

- Designed a term project for 15-821 (Mobile and Pervasive Computing)
- Met and provided guidance to project members on a regular basis

Present

Publications (Reverse Chronological Order)

- [1] **Haithem Turki**, Michael Zollhoefer, Christian Richardt, Deva Ramanan, "PyNeRF: Pyramidal Neural Radiance Fields." Under submission.
- [2] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, "Low-Bandwidth Self-Improving Transmission of Rare Training Data." To appear in MobiCom 2023.
- [3] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, "Edge-based Privacy-Sensitive Live Learning for Discovery of Training Data." To appear in NetAISys 2023.
- [4] **Haithem Turki**, Jason Y. Zhang, Francesco Ferroni, Deva Ramanan. "[SUDS: Scalable Urban Dynamic Scenes.](#)" To appear in CVPR 2023.
- [5] Mahadev Satyanarayanan, Ziqiang Feng, Shilpa George, Jan Harkes, Roger Iyengar, **Haithem Turki**, Padmanabhan Pillai, "[Accelerating Silent Witness Storage.](#)" in IEEE Micro, Nov.-Dec. 2022.
- [6] **Haithem Turki**, Deva Ramanan, Mahadev Satyanarayanan, "[Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs.](#)" Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.
- [7] Shilpa George, Thomas Eiszler, Roger Iyengar, **Haithem Turki**, Ziqiang Feng, Junjue Wang, Padmanabhan Pillai, Mahadev Satyanarayanan, "[OpenRTIST: End-to-End Benchmarking for Edge Computing.](#)" in IEEE Pervasive Computing, vol. 19, no. 4, pp. 10-18, 1 Oct.-Dec. 2020
- [8] Mahadev Satyanarayanan, Thomas Eiszler, Jan Harkes, **Haithem Turki** and Ziqiang Feng, "[Edge Computing for Legacy Applications.](#)" in IEEE Pervasive Computing, vol. 19, no. 4, pp. 19-28, 1 Oct.-Dec. 2020
- [9] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, "[Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation.](#)" in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 37, no. 7, pp. 1373-1386, 1 July 2015.
- [10] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, "[Learning specific-class segmentation from diverse data.](#)" 2011 International Conference on Computer Vision.

Patents

- [1] **Haithem Turki**, Robert Fink, Amr Al Mallah. "[Systems and Methods for Indexing and Searching.](#)" US Patent 2019/0347343 A1, filed June 8, 2018, and issued November 14, 2019.
- [2] **Haithem Turki**, Sander Kromwijk, Stephen Cohen, Yixun Xu, Feridun Arda Kara. "[Systems and Methods for Constraint Driven Database Searching.](#)" US Patent 2018/293239 A1, filed April 11, 2017, and issued October 11, 2018.

Technical Reports

- [1] Ziqiang Feng, Shilpa George, **Haithem Turki**, Roger Iyengar, Padmanabhan Pillai, Jan Harkes, Mahadev Satyanarayanan. [“Improving Edge Elasticity via Decode Offload.”](#) CMU-CS-21-139, September 2021.

Additional Information

- **Conference Reviewer:** CVPR, ICCV, SIGGRAPH Asia
- **Volunteer Activities:** Alchemist Accelerator (mentor and investor)
- **Languages:** Fluent in French and Arabic. Currently learning Japanese and German.
- **Other Qualifications:** Eberly Future Faculty Program. Top Secret security clearance (current). FINRA Series 65. Graduate of the Stanford StartX startup accelerator.